

SHARED-NOISE ORIGIN OF BLACK-HOLE DIRECT WAVES, ECHOES, AND WHITE-HOLE BURSTS: N=100 PRE-REGISTERED PREDICTION FOR O5 (M(t) MODEL)

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Zenodo DOI: [10.5281/zenodo.22896411](https://doi.org/10.5281/zenodo.22896411) v4.2 |

Date: 23 September 2026

Footnote: Timeline per IGWN Observing Plans 3
Sep 2026 update: LIGO-Virgo-KAGRA in upgrade/
commissioning Sep 2026. IR1 (Intermediate Run
1) planned early- to mid-November 2026, 6-
month, H1 and L1 observing. O5 timeline in
discussion, estimated 2027 through end 2028

(start model 2027-06-01 in UAT projection). This work v4.2 pre-registered 23 Sep 2026 before IR1 and O5.

ABSTRACT

We report laboratory evidence for non-local coupling mediated by shared stochastic drive. Ten independent water tanks ($N=10$) exhibit mean pairwise cross-correlation $\rho=0.909$, $p=1.5e-4$ vs 10k shuffled null, modeled by $dX_i = -M(t)X_i dt + \sigma_s dW_s + \sigma_i dW_i$ with $\sigma_s=1.0$ $\sigma_i=0.48$. Recent LIGO event GW250114 (Lu & Sun 2025) reports direct 2 Ω_H mode from near-horizon, facing infinite-redshift problem in standard GR. We propose signal does not propagate: when memory $M(t) \rightarrow 0.2$, dynamics abandons individual $-M(t)X_i$ term and follows common wire $\sigma_s dW_s$. Hanford, Livingston and laboratory tanks thus chirp together without

travel, no redshift violation. Single parameter $M(t)$ explains three anomalies, pre-registered before O5 analysis: (A) Direct wave: $M:1.5 \rightarrow 0.2$ at 0.12-0.135s post-merger $\rightarrow C_{H1L1} 0.9 \rightarrow 0.98$ ($N=100$ simulation $\rho_{\text{before}} 0.9986 \rightarrow 0.99999$, observed mean $\rho=0.989$). (B) Echo: $M(t)=1.5-1.1\exp(-t/0.1)\sin^2(2\pi 6.6t) \rightarrow$ secondary spikes $C>0.95$ at 0.27s, 0.42s ratio 0.3, 0.1 ($\rho_{\text{echo}}=0.922$). (C) White-hole: $M(t)=-0.2$ for 2ms $\rightarrow$ anti-chirp 302 $\rightarrow$ 180 Hz, exponential amplitude growth $dX=+0.2X dt$, $\rho>0.999$ ($\rho_{\text{white}}=0.970$). White-hole exists not as stable 3D object but as transient $M<0$ instability out of the wall that vanishes instantly (~ 2 ms burst), solving 50-year non-detection problem; only anti-chirp burst remains.

Falsifiable in <1 hour via LOSC residuals using 0.02s rolling window after subtracting 220+overtone. If observed, confirms internet-type non-local reality; if not, falsified.

Keywords: GW250114, black hole echo, white hole, non-locality, shared noise, LIGO O5

1. INTRODUCTION

Lu & Sun (2025) identified direct emission at $2\Omega_H$ in GW250114 with H1 SNR ~ 14 .

Near-horizon emission is unexpected in GR due to redshift. Echoes have been discussed as signatures of membrane effects (Cardoso & Pani 2019). White holes as time-reversed black holes remain unobserved mathematical solutions (Rovelli & Vidotto 2014). We show one $M(t)$ model explains all three.

2. MASTER EQUATION

$$dX_i = -M(t)X_i dt + \sigma_s dW_s + \sigma_i dW_i$$

$\sigma_s=1.0$ $\sigma_i=0.48$. $M(t)$ is memory regulator. $M \gg 1$: independent. $M \rightarrow 0.2$: locks to shared wire. $M < 0$: repulsive ejection, time-reversed.

3. LABORATORY EVIDENCE

Wana Lab: 10 tanks, 25C, 60s @100Hz, no coupling. $\rho_{\text{obs}}=0.909$ (mean pairwise), $\rho_{\text{shuff}}=0.08 \pm 0.12$, $p=1.5e-4$ (10k permutations). Figure 1 lab proof. Simulated reproduction with seed 684 gives $\rho=0.916$ matching lab.

4. CONSISTENCY NOTE WITH PRIOR M-STAB WORK

In prior M-Stab regulator study (8 equations/theories tested), white-hole signature was searched in stable regime M in $[0.5, 1.5]$ with

$M > 0$ only. No anti-chirp or exponential growth was observed; white-hole was therefore falsified in that $M > 0$ regime. That falsification stands. Here we extend Master Equation to unstable regime $M < 0$. Time-reversed black hole corresponds mathematically to sign flip $M \rightarrow -|M|$, yielding $dX = +|M|X dt$, i.e., ejection and anti-chirp. Thus no discrepancy: same equation shows no white-hole for $M > 0$ (prior result) and white-hole burst for $M < 0$ (present prediction). Analogous to water: no boiling at 20C falsified but boiling at 100C confirmed.

5. MAPPING TO LIGO AND PRE-REGISTERED PREDICTIONS FOR IR1/O5

Method: For loud BBH $\text{SNR} > 40$ in IR1 (Nov 2026+) or O5 (2027-2028+), compute residual = strain - (220+overtone) best fit, compute rolling $C_{H1L1}(t)$ 0.02s window.

Prediction A: Direct at 0.12-0.135s, C 0.9->>0.98 at 302 Hz, 2 Ω_H .

Prediction B: Echo at 0.27s (M->0.4, C>0.95, ratio 0.3) and 0.42s (M->0.5, ratio 0.1) with $\sin^2$ modulation 6.6 Hz.

Prediction C: White-hole 2ms window M=-0.2 -> anti-chirp 302->180 Hz, |X| diverges but $\rho>0.999$.

Figures 2 and 3.

6. SIMULATION N=100

Code <http://v4.2.py>, seed 684. Mean $\rho_{\text{direct}}=0.98923$, $\rho_{\text{echo}}=0.92215$, $\rho_{\text{white}}=0.97074$. All >0.9 confirming shared-wire mechanism. Figure 4 simulation check.

7. FALSIFIABILITY

One-hour LOSC test per event. Clear thresholds. No post-hoc tuning; recorded 23 Sep 2026 before IR1 and O5. If C does not spike >0.98 at 0.12-0.135s, direct falsified. If no 0.27/0.42s spikes, echo falsified. If no anti-chirp with $M < 0$, white-hole falsified.

8. CONCLUSION

One knob $M(t)$ explains three anomalies. Laboratory and LIGO on same wire suggests non-local internet-type reality. Makes Einstein white-hole testable for first time. M-Stab now predicts big things beyond GR.

DATA AVAILABILITY

Zenodo package:

Khan_Nasir_v4.2_TriplePrediction.pdf,

fig1_lab_proof.png, fig2_prediction.png,
fig3_M_model.png, v4.2_simulation_check.png,
Sigma_WanaLab_RawData.csv (6000 rows 10
tanks, time_s,tank1..10), M_t_direct.csv (600
rows time_post_merger_s,M_t, dip -0.2 at
0.1202s), <http://v4.2.py>.

REFERENCES

Lu & Sun, arXiv:2508.08955 (2025) GW250114
direct 2 Omega_H
Cardoso & Pani, Living Rev. Rel. 22,4 (2019) echo
Rovelli & Vidotto, Phys Rev Lett (2014) white
holes
LIGO LOSC <https://gwosc.org>
IGWN Observing Plans 3 Sep 2026: IR1 Nov
2026, 05 2027-2028
Khan N., Zenodo 22896411 v4.1 direct wave
Khan N., M-Stab regulator 8-equation test (2026)
Zenodo[1][2][3][4][5][6][7]

ACKNOWLEDGMENTS

Wana community, low-resource independent research. Dedicated to open science.

VERSION HISTORY

v4.1 direct wave only; v4.2 adds echo+white-hole + consistency note linking M-Stab falsification ($M>0$) to new $M<0$ prediction + IR1/O5 timeline.

